Student ID: 6788183
Operating system: 64-bit operating system
Docker context: default Current DOCKER_HOST based configuration
npipe:////./pipe/docker_engine
Repository: https://github.com/Yumon-lab/itcs258-691-lab11-Dockerization-for-Production

1) preflight

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production> node --version
v24.19.0
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production> python --version
Python 3.12.10
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production> docker --version
Docker version 29.8.0, build 88096ef
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production> docker info
Client:
 Version:    29.8.0
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.140.0
    Path:     C:\Users\user\.docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.31.0
    Path:     C:\Users\user\.docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)

Server:
 Containers: 4
  Running: 1
  Paused: 0
  Stopped: 3
 Images: 8
 Server Version: 29.8.0
 Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
 Logging Driver: json-file
 Cgroup Driver: cgroupfs
 Cgroup Version: 2
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api> d
ocker run --rm hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
```

Before we start our docker application, we have to check node and python version as these two are crucial for development, we must also check docker version to see Docker client where we issue docker command and docker server that manipulate image. We then run docker info to reveal the information of docker engine.
docker run --rm hello-world we run this to test whether docker client can communicate with the engine

2) express demo

```
>>
{"status":"ok","framework":"express"}
PS C:\Users\user\Dcurl.exe http://localhost:3000/      nt\Week 11\itcs258-691-lab11-Dockerizatio
{"message":"Hello from local Express","lab":11}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production> curl.exe http://localhost:3000/health
>>
{"status":"ok","framework":"express"}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production>
```

Do you want to install the recommended 'Co
extension from Microsoft for the Docker lang
Install

/ endpoint and /health endpoint answer

projects > express-api > Dockerfile

```
1   FROM node:24-bookworm-slim
2   WORKDIR /app
3   COPY package*.json ./
4   RUN npm ci --omit=dev
5   COPY server.js ./
6   ENV NODE_ENV=production
7   ENV PORT=3000
8   USER node
9   EXPOSE 3000
10  CMD ["node", "server.js"]
11
```

Dockerfile

lab11-instructions.md | server.js 1, M | report-template.md | .dockerignore

projects > express-api > .dockerignore

```
1   node_modules
2   .git
3   .env
4   .env.*
5   !.env.example
6   *.log
7
```

.dockerignore

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker build -t lab11-express:v1 .
[+] Building 2.4s (11/11) FINISHED                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                0.0s
 => => transferring dockerfile: 227B                                                0.0s
 => [internal] load metadata for docker.io/library/node:24-bookworm-slim            2.0s
 => [auth] library/node:pull token for registry-1.docker.io                         0.0s
 => [internal] load .dockerignore                                                   0.0s
 => => transferring context: 90B                                                    0.0s
 => [1/5] FROM docker.io/library/node:24-bookworm-slim@sha256:c86ecead9e55a5210ac1727de38  0.0s
 => => resolve docker.io/library/node:24-bookworm-slim@sha256:c86ecead9e55a5210ac1727de38  0.0s
 => [internal] load build context                                                   0.1s
 => => transferring context: 30.43kB                                                0.0s
 => CACHED [2/5] WORKDIR /app                                                       0.0s
 => CACHED [3/5] COPY package*.json ./                                              0.0s
 => CACHED [4/5] RUN npm ci --omit=dev                                              0.0s
 => CACHED [5/5] COPY server.js ./                                                  0.0s
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker run -d --name lab11-express -p 127.0.0.1:8080:3000
 --env-file .env.example lab11-express:v1
6b957a07dd64ba97abda85852cde57ffa7a73aeeeeed1836447e43ac7bcee927
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED         STATUS         PORTS
                NAMES
6b957a07dd64   lab11-express:v1   "docker-entrypoint.s…"   5 seconds ago   Up 4 seconds   127.0.
0.1:8080->3000/tcp   lab11-express
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker logs lab11-express
Express listening on port 3000
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> curl.exe http://localhost:8080/
{"message":"Hello from Docker Express","lab":11}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> curl.exe http://localhost:8080/health
{"status":"ok","framework":"express"}
```

image build; docker run, docker ps, logs and 8080:3000 mapping;

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker exec lab11-express node -e "fetch('http://127.0.0.
1:3000/health').then(r=>r.json()).then(console.log)"
{ status: 'ok', framework: 'express' }
```

Verify from inside

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker stop lab11-express
lab11-express
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker rm lab11-express
lab11-express
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker run -d --name lab11-express -p 127.0.0.1:8080:3000
 -e APP_MESSAGE="Hello Student 1234567" lab11-express:v1
d717cfe2d7fb4fea2837a2b055c4ee301200c076e0bbf20fbd56060d82bdc352
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> curl.exe http://localhost:8080/
{"message":"Hello Student 1234567","lab":11}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api>
```

Change configuration without rebuilding and express container

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker build -t lab11-express:v2 .
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker stop lab11-express
>>
lab11-express
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker rm lab11-express
lab11-express
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker run -d --name lab11-express -p 127.0.0.1:8080:3000
 --env-file .env.example lab11-express:v2
a0c444011a339bfb1dbecfcced9307de78f983e130836c82bf97a297f881f454
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> curl.exe http://localhost:8080/
>>
{"message":"Hello from Docker Express","lab":111}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api>
```

Show the greeting change using a runtime variable, then the lab: 111 result after editing source, rebuilding and recreating the container, we do this because we must remove the old one and replace with the new container we have created

```
n-for-Production\projects\express-api> docker stop lab11-express
lab11-express
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker rm lab11-express
lab11-express
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerizatio
n-for-Production\projects\express-api>
```

Stop and Removed the lab container after finished

```
[notice] To update, run: C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fastapi-api\.venv\Scripts\python.exe -m pip install --upgrade pip

C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fastapi-api>.\.venv\Scripts\python.exe main.py
INFO:     Started server process [14904]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:     127.0.0.1:64900 - "GET / HTTP/1.1" 200 OK
INFO:     127.0.0.1:64902 - "GET /health HTTP/1.1" 200 OK
```

3)FastAPI demo

```
Problems  Output  Debug Console  Ports  Terminal  SonarQube

PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production> curl.exe http://localhost:8000/
{"message":"Hello from local FastAPI","lab":111}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production> curl.exe http://localhost:8000/health
{"status":"ok","framework":"fastapi"}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production>
```

Show the local Python environment, local and container API responses

lab11-instructions.md | Dockerfile 1

projects > express-api > fastapi-api > Dockerfile

```
1   FROM python:3.12-slim
2   WORKDIR /app
3   ENV PYTHONDONTWRITEBYTECODE=1
4   ENV PYTHONUNBUFFERED=1
5   ENV PORT=8000
6   COPY requirements.txt ./
7   RUN pip install --no-cache-dir -r requirements.txt
8   RUN useradd --create-home --uid 10001 appuser
9   COPY main.py ./
10  USER appuser
11  EXPOSE 8000
12  CMD ["python", "main.py"]
```

Dockerfile

lab11-instructions.md | .dockerignore

projects > express-api > fastapi-api > .dockerignore

```
1   .venv
2   __pycache__
3   *.__pycache__
4   .git
5   .env
6   .env.*
7   ! .env.example
```

.dockerignore

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker build -t lab11-fastapi:v1 .
 => => exporting manifest sha256:22d68286876346fb2ba994e657217f6da1fbdcef7d6d1628d4cdb3aded23435e                                0.0s
 => => exporting config sha256:913c128ae57ceebf5b3c614c6bf0c4ab88e36d256b5ffd3db0070b3206deeb94                                  0.0s
 => => exporting attestation manifest sha256:96e4925c006a24cd9db7324ba1f9385ceaaefe67fdb08036cb5fbe82b5938a4e                    0.0s
 => => exporting manifest list sha256:80422e9ebabc6bf7adfdbd7b614665b9bf43614d1f63b9a19f1dc9e8471a07cc                           0.0s
 => => naming to docker.io/library/lab11-fastapi:v1                                                                              0.0s
 => => unpacking to docker.io/library/lab11-fastapi:v1                                                                           0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/5hjabsvt7gpsf2ce8fph740az
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker run -d --name lab11-fastapi -p 127.0.0.1:8081:8000 --env-file .env.example lab11-fastapi:v1
e896713afe26cd26198cf62b83a828010399002ad6c59912ed85aacdc82046bb
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker logs lab11-fastapi
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> curl.exe http://localhost:8081/
{"message":"Hello from Docker FastAPI","lab":11}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> curl.exe http://localhost:8081/health
{"status":"ok","framework":"fastapi"}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api>
```

image build, running container, logs,8081:8000.

```
PS C:\Usercurl.exe http://localhost:8081/^C
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker exec lab11-fastapi python -c "import urllib.request; print(urllib.request.urlopen('http://127.0.0.1:8000/health').read()
.decode())"
{"status":"ok","framework":"fastapi"}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api>
```

Host and inside container requests

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker stop lab11-fastapi
lab11-fastapi
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker rm lab11-fastapi
lab11-fastapi
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker run -d --name lab11-fastapi -p 127.0.0.1:8081:8000 -e APP_MESSAGE="Python configuration changed" lab11-fastapi:v1
73cacb3ea16d0e5e5498196db40152ce047e08008b742e6cf97dc49ff5df0736
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> curl.exe http://localhost:8081/
{"message":"Python configuration changed","lab":11}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api>
```

Configuration change

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker build -t lab11-fastapi:v2 .
 => CACHED [3/6] COPY requirements.txt ./                                                                                        0.0s
 => CACHED [4/6] RUN pip install --no-cache-dir -r requirements.txt                                                              0.0s
 => CACHED [5/6] RUN useradd --create-home --uid 10001 appuser                                                                   0.0s
 => CACHED [6/6] COPY main.py ./                                                                                                 0.0s
 => exporting to image                                                                                                           0.1s
 => => exporting layers                                                                                                          0.0s
 => => exporting manifest sha256:dc7ec9fbb4efcb138e6262698e2777a3ccd470fbc374614b4fdc1de8f83043d0                                0.0s
 => => exporting config sha256:50a0ca595ed8f86beb0a5d1a00e63144c8cbbaa51b32b139b2c030b743768fff                                  0.0s
 => => exporting attestation manifest sha256:075ee8a6a8d56587699e9a71c222e1ca1250d6499ea5e9ec9f303ce6a21e5907                    0.0s
 => => exporting manifest list sha256:b6704fe0b81302a74b5f2002000a5a49a37b5eb938d132745c06f3392c08c017                           0.0s
 => => naming to docker.io/library/lab11-fastapi:v2                                                                              0.0s
 => => unpacking to docker.io/library/lab11-fastapi:v2                                                                           0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/tcetnboqwvpfuwg1k3qje5djx
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker stop lab11-fastapi
lab11-fastapi
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker rm lab11-fastapi
lab11-fastapi
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker run -d --name lab11-fastapi -p 127.0.0.1:8081:8000 --env-file .env.example lab11-fastapi:v2
bb50f52d9c7104bf9f4458ef11ba1a412dd009c0c7e8a91ce79fdedae35b1c63
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> curl.exe http://localhost:8081/
{"message":"Hello from Docker FastAPI","lab":111}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api>
```

source edit/rebuild

```
tapi-api> docker stop lab11-fastapi
lab11-fastapi
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker rm lab11-fastapi.
Error response from daemon: No such container: lab11-fastapi.
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\projects\express-api\fas
tapi-api> docker rm lab11-fastapi .
lab11-fastapi
Error response from daemon: page not found
```

4)NestJS bridge

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> npm run build

> itcs258-lab11-api@1.0.0 build
> tsc

PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> npm start

> itcs258-lab11-api@1.0.0 start
> node dist/main.js

[Nest] 29956  - 11/02/2026, 1:37:53 AM     LOG [NestFactory] Starting Nest application...
[Nest] 29956  - 11/02/2026, 1:37:53 AM     LOG [InstanceLoader] AppModule dependencies initialized +6ms
[Nest] 29956  - 11/02/2026, 1:37:53 AM     LOG [RoutesResolver] AppController {/}: +14ms
[Nest] 29956  - 11/02/2026, 1:37:53 AM     LOG [RouterExplorer] Mapped {/, GET} route +2ms
[Nest] 29956  - 11/02/2026, 1:37:53 AM     LOG [RouterExplorer] Mapped {/health, GET} route +0ms

PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production> curl.exe http://localho
st:3000/
>>
{"message":"Hello from local NestJS","lab":11}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production> curl.exe http://localho
st:3000/health
{"status":"ok","environment":"development"}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production>
```

Typescript build and Local requests

lab11-instructions.md | Dockerfile

lab11-api > Dockerfile

```dockerfile
# Compile TypeScript with development dependencies.
FROM node:24-bookworm-slim AS build
WORKDIR /app
COPY package*.json ./
RUN npm ci
COPY tsconfig.json ./
COPY src ./src
RUN npm run build

# Keep only runtime dependencies and compiled JavaScript.
FROM node:24-bookworm-slim AS runtime
WORKDIR /app
ENV NODE_ENV=production
ENV PORT=3000
COPY package*.json ./
RUN npm ci --omit=dev && npm cache clean --force
COPY --from=build --chown=node:node /app/dist ./dist
USER node
EXPOSE 3000
CMD ["node", "dist/main.js"]
```

Dockerfile

lab11-instructions.md | .dockerignore

lab11-api > .dockerignore

```
node_modules
dist
.git
.env
.env.*
!.env.example
*.log
coverage
```

.dockerignore

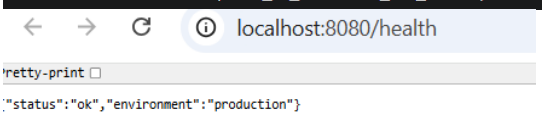
```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker build
-t itcs258-api:lab11 .
[+] Building 2.3s (15/15) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                              0.0s
 => => transferring dockerfile: 551B                                                                                              0.0s
 => [internal] load metadata for docker.io/library/node:24-bookworm-slim                                                          1.9s
 => [auth] library/node:pull token for registry-1.docker.io                                                                       0.0s
 => [internal] load .dockerignore                                                                                                 0.0s
 => => transferring context: 104B                                                                                                 0.0s
 => [internal] load build context                                                                                                 0.0s
 => => transferring context: 48.66kB                                                                                              0.0s
 => [build 1/7] FROM docker.io/library/node:24-bookworm-slim@sha256:c86ecead9e55a5210ac1727de38a4ca840bf03dcaf34dc9749addbf383b192  0.0s
```

Image build

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker run -d
 --name lab11-api -p 127.0.0.1:8080:3000 --env-file .env.example itcs258-api:lab11
d4d067989875f8615fe3e004460e32debc7127d2ec9cda13886d1ebc3aca19f
```

Running container

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED         STATUS         PORTS                    NAMES
d4d067989875   itcs258-api:lab11   "docker-entrypoint.s…"   3 minutes ago   Up 3 minutes   127.0.0.1:8080->3000/tcp   lab11-api
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker logs l
ab11-api
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [NestFactory] Starting Nest application...
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [InstanceLoader] AppModule dependencies initialized +8ms
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [RoutesResolver] AppController {/}: +3ms
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [RouterExplorer] Mapped {/, GET} route +3ms
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [RouterExplorer] Mapped {/health, GET} route +1ms
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [NestApplication] Nest application successfully started +2ms
Lab 11 API listening on port 3000
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> curl.exe http
://localhost:8080/
{"message":"Hello from Docker","lab":11}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> curl.exe http
://localhost:8080/health
{"status":"ok","environment":"production"}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api>
```



Logs and port mapping

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker exec l
ab11-api node -e "fetch('http://127.0.0.1:3000/health').then(r=>r.json()).then(console.log)"
{ status: 'ok', environment: 'production' }
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api>
```

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker exec l
ab11-api node -e "fetch('http://127.0.0.1:3000/health').then(r=>r.json()).then(console.log)"
{ status: 'ok', environment: 'production' }
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker logs -
-tail 30 lab11-api
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [NestFactory] Starting Nest application...
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [InstanceLoader] AppModule dependencies initialized +8ms
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [RoutesResolver] AppController {/}: +3ms
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [RouterExplorer] Mapped {/, GET} route +3ms
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [RouterExplorer] Mapped {/health, GET} route +1ms
[Nest] 1  - 11/01/2026, 6:45:39 PM     LOG [NestApplication] Nest application successfully started +2ms
Lab 11 API listening on port 3000
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker inspec
t --format "{{.Config.User}}" lab11-api
node
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker inspec
t --format "{{json .NetworkSettings.Ports}}" lab11-api
{"3000/tcp":[{"HostIp":"127.0.0.1","HostPort":"8080"}]}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api>
```

Inside-container requests

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker stop lab11-api
lab11-api
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker rm lab11-api
lab11-api
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker run -d --name lab11-api -p 127.0.0.
1:8080:3000 -e PORT=3000 -e APP_MESSAGE="Hello Student 1234567" itcs258-api:lab11
7050af31d53984a3e25d567390f1ae17bf32e5fbeea777a5508cd061b9366462
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker logs lab11-api
[Nest] 1  - 11/01/2026, 6:54:05 PM     LOG [NestFactory] Starting Nest application...
[Nest] 1  - 11/01/2026, 6:54:05 PM     LOG [InstanceLoader] AppModule dependencies initialized +4ms
[Nest] 1  - 11/01/2026, 6:54:05 PM     LOG [RoutesResolver] AppController {/}: +3ms
[Nest] 1  - 11/01/2026, 6:54:05 PM     LOG [RouterExplorer] Mapped {/, GET} route +1ms
[Nest] 1  - 11/01/2026, 6:54:05 PM     LOG [RouterExplorer] Mapped {/health, GET} route +1ms
[Nest] 1  - 11/01/2026, 6:54:05 PM     LOG [NestApplication] Nest application successfully started +1ms
Lab 11 API listening on port 3000
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> curl.exe http://localhost:8080/
{"message":"Hello Student 1234567","lab":11}
```

Experiment A · change the greeting without rebuilding

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker build -t itcs258-api:lab11-v2 .
 => => exporting config sha256:c70ea22afb979de96c5169c22a2ee63d82bea23a2871fd84bd45bc1229dcd5e4                                                    0.0s
 => => exporting attestation manifest sha256:befedfedf38306e31e253cfcf6aa7b755f97d6e742da2b34581baf2021f01ee7                                      0.0s
 => => exporting manifest list sha256:56fe577a8e8a0d27ca9994c297f96c60ae4a7dbeef4f49149e44f7beae74f08e                                             0.0s
 => => naming to docker.io/library/itcs258-api:lab11-v2                                                                                           0.0s
 => => unpacking to docker.io/library/itcs258-api:lab11-v2                                                                                        0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/76dcqleejfzht2auavxfy0b99
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker stop lab11-api
lab11-api
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker rm lab11-api
>>
lab11-api
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker run -d --name lab11-api -p 127.0.0.
1:8080:3000 --env-file .env.example itcs258-api:lab11-v2
2a8894c1c47875dcc478905710f65ebc22f3cda28c4d5c82d9423edac5134b11
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> curl.exe http://localhost:8080/
>>
{"message":"Hello from Docker","lab":111}
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api>
```

Experiment B · change the code and rebuild

```
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker stop lab11-api
lab11-api
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker rm lab11-api
lab11-api
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED       STATUS                     PORTS                    NAMES
ed53b1368616   redis     "docker-entrypoint.s…"   4 weeks ago   Exited (255) 2 days ago   0.0.0.0:6379->6379/tcp   redis
PS C:\Users\user\Desktop\ICT_MU_work\Back_End_development\Week 11\itcs258-691-lab11-Dockerization-for-Production\lab11-api>
```

clean up